

Agile projects commonly use Sprint Planning, Story Points, Velocity, Burndown Charts, Capacity, and Earned Value for numerical analysis.

In simple words

Project = Entire journey

Sprint = One short development cycle within that journey

For example:

Project → 6 months

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Sprint 1 → 2 weeks

Sprint 2 → 2 weeks

Sprint 3 → 2 weeks

Sprint 4 → 2 weeks

...and so on until the project objectives are achieved.

Question 1: Calculate Sprint Velocity

A Scrum team completes the following User Stories during three sprints:

Sprint	Story Points Completed
Sprint 1	24
Sprint 2	30
Sprint 3	27

Calculate the average velocity of the team.

Answer

Formula:

$$\begin{aligned} \text{Average Velocity} &= \frac{\text{Total Story Points}}{\text{Number of Sprints}} \\ &= \frac{24 + 30 + 27}{3} \\ &= \frac{81}{3} \\ &= \boxed{27 \text{ Story Points/Sprint}} \end{aligned}$$

Interpretation: The team can reasonably be expected to complete approximately 27 story points per sprint.

Question 2: Calculate Number of Sprints Required

An Agile project has a product backlog containing 150 story points. The team's average velocity is 25 story points per sprint. Each sprint lasts 2 weeks.

Calculate:

1. Number of sprints required
2. Approximate project duration

Formula:

$$\begin{aligned} \text{Number of Sprints} &= \frac{\text{Total Story Points}}{\text{Velocity}} \\ &= \frac{150}{25} \\ &= \boxed{6 \text{ Sprints}} \end{aligned}$$

Each sprint = 2 weeks.

Therefore:

$$\begin{aligned} \text{Project Duration} &= 6 \times 2 \\ &= \boxed{12 \text{ weeks}} \end{aligned}$$

Final Answer

- Sprints required = 6
- Project duration = 12 weeks

Question 3: Agile Team Capacity

An Agile development team has 5 members. Each member is available for 6 productive hours per day. The sprint duration is 10 working days. However, the team expects 20% of its time to be consumed by meetings, administrative activities and other non-development work. Calculate the team's effective sprint capacity.

Step 1: Total available hours

$$\begin{aligned} &5 \times 6 \times 10 \\ &= 300 \text{ hours} \end{aligned}$$

Step 2: Non-development time

$$20\% \times 300 = 60 \text{ hours}$$

Step 3: Effective capacity

$$300 - 60 = 240 \text{ hours}$$

240 hours

Answer: Effective Sprint Capacity = 240 hours.

Question 4: Calculate Agile Project Completion Using Velocity

A product backlog contains 240 story points.

The team has completed:

- Sprint 1 = 35 points
- Sprint 2 = 40 points
- Sprint 3 = 45 points
- Sprint 4 = 40 points

Calculate:

1. Average velocity
2. Remaining story points
3. Estimated number of additional sprints required

Step 1: Total completed points

$$35 + 40 + 45 + 40 = 160$$

Step 2: Average velocity

$$\begin{aligned} \text{Velocity} &= \frac{160}{4} \\ &= \boxed{40 \text{ points/sprint}} \end{aligned}$$

Step 3: Remaining backlog

$$240 - 160 = 80$$

Therefore:

$$\boxed{80 \text{ points remaining}}$$

Step 4: Additional sprints

$$\begin{aligned} \frac{80}{40} &= 2 \\ &\boxed{2 \text{ sprints}} \end{aligned}$$

Measure	Answer 
Average velocity	40 points/sprint
Completed points	160
Remaining points	80
Additional sprints	2

Question 5: Sprint Burndown Calculation

A team starts a sprint with 100 story points.

During the sprint, the team completes:

Day	Points Completed
Day 1	8
Day 2	12
Day 3	10
Day 4	15
Day 5	10

Calculate:

1. Total points completed
2. Remaining backlog after Day 5
3. Average daily completion rate



Step 1: Total completed

$$8 + 12 + 10 + 15 + 10 = 55$$

55 points

Step 2: Remaining backlog

$$100 - 55 = 45$$

45 points

Step 3: Average completion rate

$$\frac{55}{5} = 11$$

11 points/day

Interpretation: The team has completed 55% of the sprint backlog and has 45 points remaining.

Question 6: Calculate Agile Project Cost

An Agile team consists of:

- 4 developers earning 2,000 per day each
- 1 Scrum Master earning 3,000 per day
- 1 Product Owner earning 3,500 per day

The project requires 10 sprints, and each sprint consists of 10 working days.

Calculate the total personnel cost.

Step 1: Daily developer cost

$$4 \times ₹2,000 = ₹8,000$$

Step 2: Total daily team cost

$$\begin{aligned} ₹8,000 + ₹3,000 + ₹3,500 \\ = ₹14,500 \end{aligned}$$

Step 3: Total project working days

$$10 \times 10 = 100 \text{ days}$$

Step 4: Total personnel cost

$$\begin{aligned} ₹14,500 \times 100 \\ = \boxed{₹14,50,000} \end{aligned}$$

Answer: Total personnel cost = ₹14.50 lakh.